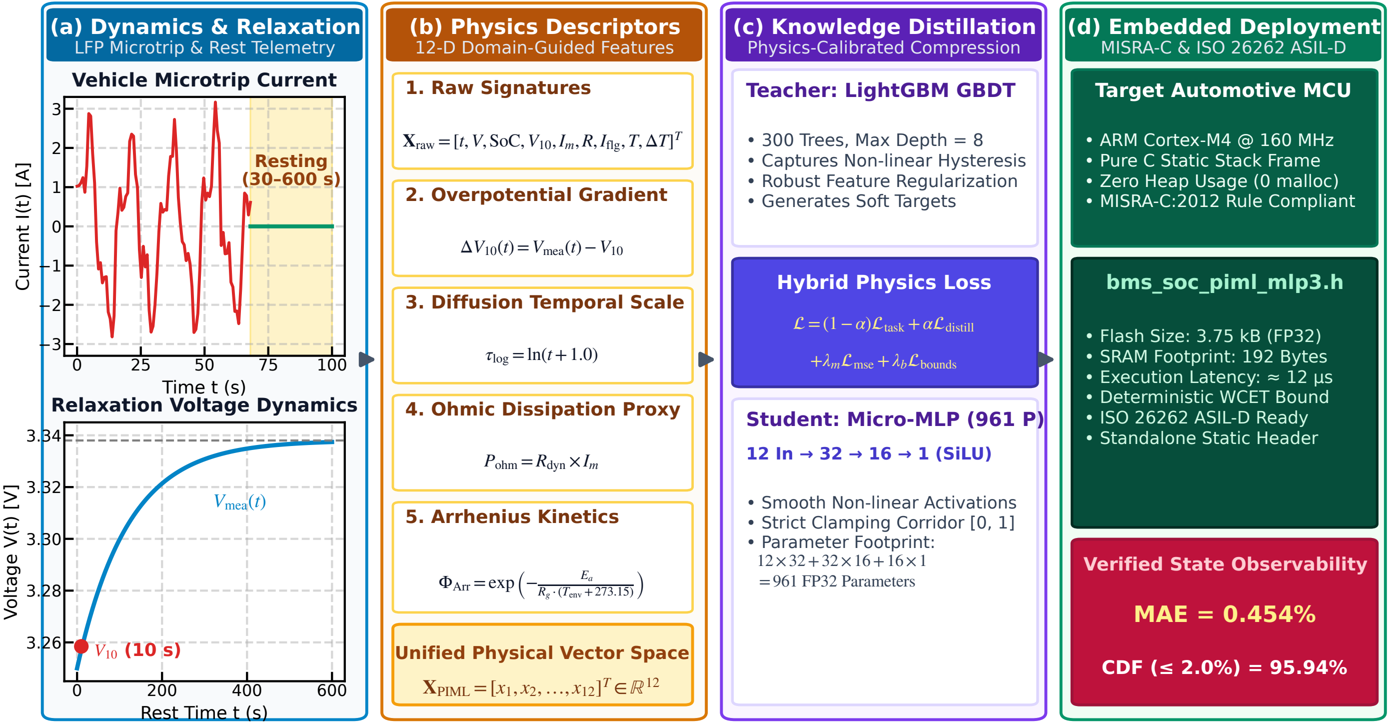
(a) Dynamics & Relaxation
LFP Microtrip & Rest Telemetry
Vehicle Microtrip Current
Current I(t) [A]
Resting (30-600 s)
Time t (s)
Relaxation Voltage Dynamics
Voltage V(t) [V]
$V_{\mathrm{mea}}(t)$
V_{10} (10 s)
Rest Time t (s)
(b) Physics Descriptors
12-D Domain-Guided Features
1. Raw Signatures
$\mathbf{X}_{\mathrm{raw}} = [t, V, \mathrm{SoC}, V_{10}, I_m, R, I_{\mathrm{flg}}, T, \Delta T]^T$
2. Overpotential Gradient
$\Delta V_{10}(t) = V_{\mathrm{mea}}(t) - V_{10}$
3. Diffusion Temporal Scale
$\tau_{\log} = \ln(t + 1.0)$
4. Ohmic Dissipation Proxy
$P_{\mathrm{ohm}} = R_{\mathrm{dyn}} \times I_m$
5. Arrhenius Kinetics
$\Phi_{\mathrm{Arr}} = \exp\left(-\frac{E_a}{R_g \cdot (T_{\mathrm{env}} + 273.15)}\right)$
Unified Physical Vector Space
$\mathbf{X}_{\mathrm{PIML}} = [x_1, x_2, \ldots, x_{12}]^T \in \mathbb{R}^{12}$
(c) Knowledge Distillation
Physics-Calibrated Compression
Teacher: LightGBM GBDT
• 300 Trees, Max Depth = 8
• Captures Non-linear Hysteresis
• Robust Feature Regularization
• Generates Soft Targets
Hybrid Physics Loss
$\mathcal{L} = (1-\alpha)\mathcal{L}_{\mathrm{task}} + \alpha\mathcal{L}_{\mathrm{distill}}$
$+\lambda_m\mathcal{L}_{\mathrm{mse}} + \lambda_b\mathcal{L}_{\mathrm{bounds}}$
Student: Micro-MLP (961 P)
12 In → 32 → 16 → 1 (SiLU)
• Smooth Non-linear Activations
• Strict Clamping Corridor [0, 1]
• Parameter Footprint:
$12 \times 32 + 32 \times 16 + 16 \times 1$
$= 961$ FP32 Parameters
(d) Embedded Deployment
MISRA-C & ISO 26262 ASIL-D
Target Automotive MCU
• ARM Cortex-M4 @ 160 MHz
• Pure C Static Stack Frame
• Zero Heap Usage (0 malloc)
• MISRA-C:2012 Rule Compliant
bms_soc_piml_mlp3.h
• Flash Size: 3.75 kB (FP32)
• SRAM Footprint: 192 Bytes
• Execution Latency: ≈ 12 μs
• Deterministic WCET Bound
• ISO 26262 ASIL-D Ready
• Standalone Static Header
Verified State Observability
MAE = 0.454%
CDF (≤ 2.0%) = 95.94%